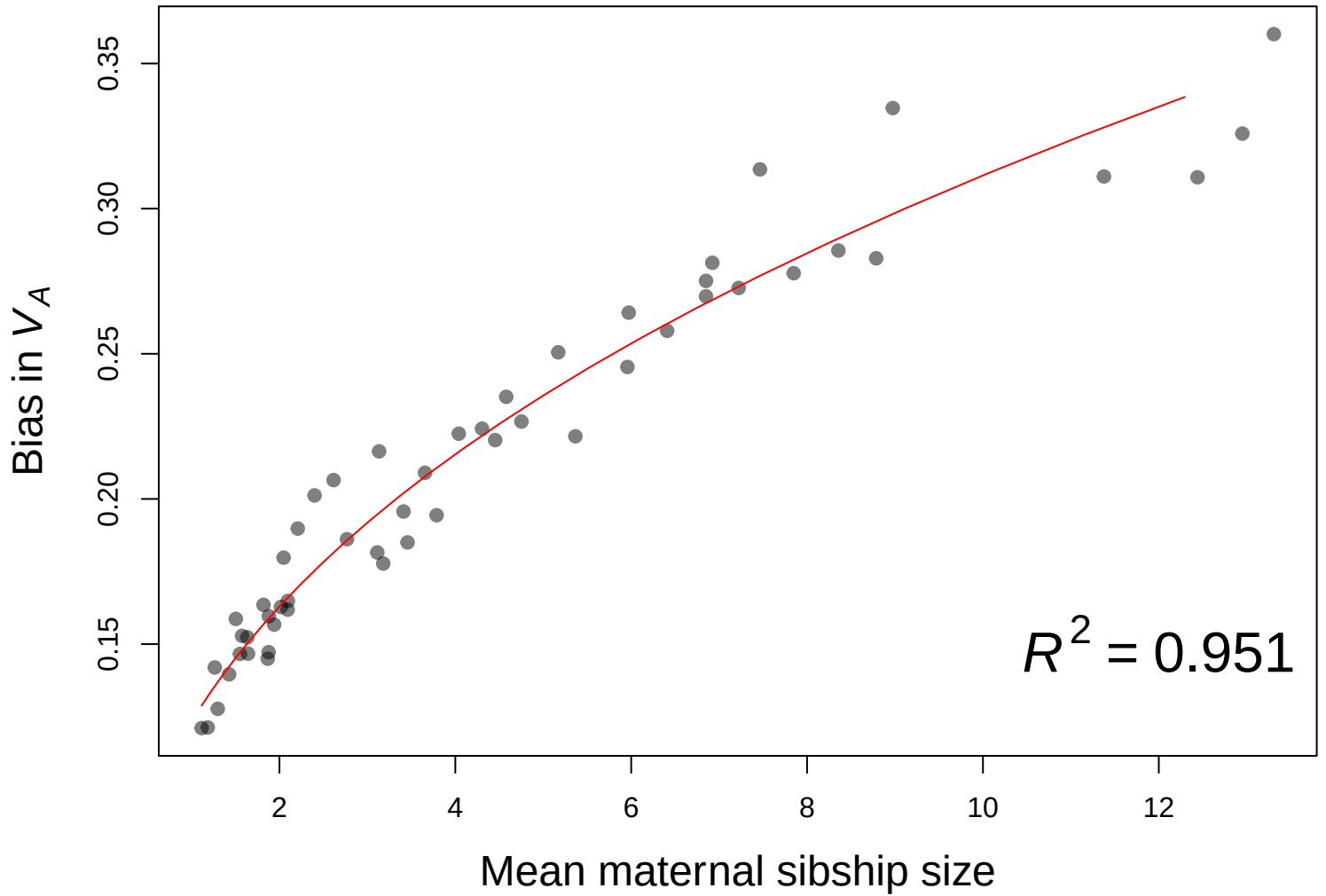
a)

Metric	R^2
records	0.202
maternities	0.374
paternities	0.316
full sibs	0.528
maternal sibs	0.731
maternal half sibs	0.685
paternal sibs	0.676
paternal half sibs	0.644
maternal grandmothers	0.176
maternal grandfathers	0.168
paternal grandmothers	0.138
paternal grandfathers	0.121
maximum pedigree depth	−0.018
founders	−0.011
mean maternal sibship size	0.951
mean paternal sibship size	0.866
non−zero F	0.082
F > 0.125	0.215
mean pairwise relatedness	0.013
pairwise relatedness>=0.125	0.021
pairwise relatedness>=0.25	0.008
pairwise relatedness>=0.5	−0.017
var pairwise relatedness	−0.003

b)